

An exact null for the Rasch change index

Why the RCI critical value is not 1.96, and how to compute it without simulating

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Purpose

`RMpersonChange()` includes an option for `critical = "simulate"` rather than the conventional 1.96, because the Rasch change index

$$RCI = \frac{\hat{\theta}_2 - \hat{\theta}_1}{\sqrt{SE(\hat{\theta}_1)^2 + SE(\hat{\theta}_2)^2}}$$

is not standard normal under the null. The departure is largest on short scales and fades only slowly, so that even at 60 items with four response categories the critical value is still around 1.93. This note shows that the null does not need simulating at all: for complete data it can be enumerated exactly. It also records what the exact critical value does across test length, sample skew and targeting, and one consequence for the RCI as a statistic.

Why the null is not $N(0, 1)$

Two mechanisms, both weakening as test information grows.

Self-normalisation. $SE(\hat{\theta}_1)$ and $SE(\hat{\theta}_2)$ are deterministic functions of the realised scores, so the denominator is not independent of the numerator. A respondent who moves to an extreme score produces a large numerator and a large denominator together, and the ratio is damped. Both tails of

the null are pulled in, so the critical value falls **below** 1.96. In a t-test the denominator is independent of the numerator. Here it is not, which is why the t reference does not help either.

Discreteness. With complete data the sum score is sufficient, so $\hat{\theta}$ takes only as many values as there are scores and the null is discrete rather than continuous.

Exact enumeration

Sufficiency is what makes this tractable. $\hat{\theta}(r)$ and $SE(r)$ are deterministic lookups, the ones `RMscoreSE()` builds, and $P(r | \theta)$ follows from the Lord-Wingersky recursion. Under the null the two occasions are independent given θ , so the joint is $P(r_1 | \theta)P(r_2 | \theta)$ and the whole null is a discrete distribution over $(k + 1)^2$ score pairs. For six items with four response categories that is 361 cells.

The two ingredients do different jobs and the enumeration needs both. Sufficiency fixes the **value** of the statistic in every cell, because $\hat{\theta}$ and SE are functions of the score alone, and no probability enters that map. A quantile also needs the **weight** on each cell, which is $P(r | \theta)$, and that is not deterministic. Lord-Wingersky computes it rather than estimating it: the recursion is the elementary symmetric function identity in probability form, the same object CML estimation runs on, and it replaces an enumeration over $\prod_i (m_i + 1)$ response patterns, 4096 here and around 10^{12} at twenty items. Nothing in this note is simulated, which is what “exact” refers to. Nor is there a closed-form expression for the critical value. It is read off the enumerated distribution as a quantile.

```
# One threshold constructor for the whole note: k items, ncat response
# categories, item locations spread over [-1.2, 1.2].
mk <- function(k, ncat) {
  lapply(seq(-1.2, 1.2, length.out = k),
    function(b) b + seq(-0.9, 0.9, length.out = ncat - 1L))
}

# Lord-Wingersky: exact sum-score distribution given theta under a PCM
score_dist <- function(thr_list, theta) {
  d <- 1
  for (thr in thr_list) {
    p <- easyRasch2:::pcm_cat_probs(theta, thr)
    nd <- numeric(length(d) + length(p) - 1L)
    for (a in seq_along(d)) {
      nd[a + seq_along(p) - 1L] <- nd[a + seq_along(p) - 1L] + d[a] * p
    }
    d <- nd
  }
  d
  # index is score + 1
}

# Exact null distribution of the RCI, pooled over a set of person locations
exact_crit <- function(thr_list, thetas, alpha = 0.05) {
  steps <- vapply(thr_list, length, integer(1L))
  lut <- vapply(0:sum(steps), function(r) {
    easyRasch2:::theta_wle(
      easyRasch2:::score_pattern(r, steps), thr_list, c(-10, 10)
    )
  }, numeric(2L))
  th <- lut[1L, ]
  se <- lut[2L, ]
  rci <- outer(seq_along(th), seq_along(th),
    function(i, j) (th[j] - th[i]) / sqrt(se[i]^2 + se[j]^2))
  )
  w <- matrix(0, length(th), length(th))
}
```

```

for (t in thetas) {
  d <- score_dist(thr_list, t)
  w <- w + outer(d, d)
}
w <- w / sum(w)
o <- order(as.numeric(rci))
v <- as.numeric(rci)[o]
cw <- cumsum(as.numeric(w)[o])
c(lower = v[which(cw >= alpha / 2)[1L]],
  upper = v[which(cw >= 1 - alpha / 2)[1L]])
}

```

Validation against simulation

Run by `dev/` scratch scripts at `sim_iter` 2000 to 4000, $n = 200$, $\alpha = .05$ two-sided. The simulated and exact values agree to within Monte Carlo error in every configuration tried, including dichotomous items, items with unequal category counts, a right-skewed person distribution and a badly off-target one. Each configuration was run on its own simulated sample, so the values are not comparable with the length table below, which uses a different person distribution. The 12-item row sitting near the 20-item value there is an instance of that, not a property of test length.

configuration	simulated	exact	difference
6 items, 4 categories, symmetric theta	-1.695 / +1.695	± 1.695	0.000
6 items, 4 categories, skewed theta	-1.725 / +1.695	± 1.695	0.030
6 items, 4 categories, skewed and off target	-1.534 / +1.534	± 1.522	0.012
20 items, 2 categories, symmetric	-1.798 / +1.798	± 1.798	0.000
20 items, 2 categories, skewed	-1.798 / +1.798	± 1.798	0.000
9 items, mixed 2/3/4 categories, symmetric	-1.721 / +1.721	± 1.721	0.000
9 items, mixed 2/3/4 categories, skewed	-1.721 / +1.721	± 1.721	0.000
12 items, 4 categories, symmetric	-1.864 / +1.864	± 1.862	0.002

The exact values are perfectly symmetric. That is the expected result: under the null the two occasions are exchangeable, so $\hat{\theta}_2 - \hat{\theta}_1$ is symmetric about zero however skewed $\hat{\theta}$ itself is. The residual asymmetry in the simulated column is Monte Carlo noise. Asymmetry becomes real only when exchangeability fails, which in practice means the two occasions differing in which items were answered.

The critical value tracks test information

```

set.seed(1)
theta <- rnorm(400, 0, 1.4)
tab <- t(vapply(c(4, 6, 10, 20, 40), function(k) {
  c(items = k, exact_crit(mk(k, 4), theta)[["upper"]])

```

```
}, numeric(2L)))
kable(tab, digits = 3, col.names = c("Items", "Critical value"))
```

Items	Critical value
4	1.644
6	1.725
10	1.815
20	1.868
40	1.915

Person locations are a sample of 400 from $N(0, 1.4)$, and the number of response categories is held at four throughout. The critical value is monotone in test length, converging on 1.96. Referring the RCI to a normal null is therefore conservative, and increasingly so the shorter the scale. At six items and four categories, using 1.96 instead of 1.725 is a materially stricter test.

The convergence is slow. Extending the same setup, integrated over $N(0, 1.4)$ by quadrature rather than sampled, the value is 1.933 at 60 items, 1.949 at 200 and 1.956 at 600, and does not reach 1.96 in that range. It is within 0.03 of 1.96 from about 60 items, where the size of a 1.96 test is roughly .047 and the difference no longer matters. Dichotomous items converge more slowly and not monotonically, because of discreteness: 1.734 at 10 items, 1.720 at 20, 1.926 at 100 and 1.953 at 600. Computed in `dev/rci_crit_by_length.R`.

Response categories have to be reported alongside item count, because a six-item scale with three categories and one with seven are not comparable instruments. They do not act on the two quantities in the same way:

```
cats <- t(vapply(c(2, 3, 4, 5, 7), function(nc) {
  thr <- mk(6, nc)
  names(thr) <- paste0("I", seq_along(thr))
  d <- as.data.frame(easyRasch2::sim_partial_score(thr, theta))
  colnames(d) <- names(thr)
  rel <- RMreliabilityCurve(d, item_params = thr, output = "dataframe")
  c(categories = nc,
    critical = exact_crit(thr, theta)[["upper"]],
    marginal = attr(rel, "marginal_ratio"))
}, numeric(3L)))
kable(cats, digits = 3,
      col.names = c("Categories", "Critical value", "Marginal reliability"))
```

Categories	Critical value	Marginal reliability
2	1.546	0.615
3	1.724	0.764
4	1.725	0.829
5	1.727	0.872
7	1.779	0.906

Reliability climbs steeply with categories, from about .60 at two to about .91 at seven. The critical value does not follow it. Almost all of its movement is the step from dichotomous to polytomous, after which it is close to flat. So categories add precision without much changing how far the null lies from 1.96,

while items move both. There is a section below with further discussion on reliability and conditional reliability.

Skew and targeting

Sample skew on its own barely moves the pooled critical value: the skewed and symmetric rows above agree to the third decimal in five of seven configurations. Targeting is what matters. Pushing the same skewed distribution 2.2 logits off target, so that 39.5% of respondents are at an extreme score at one occasion or the other, drops the critical value from 1.695 to 1.522. The mechanism is the self-normalisation above: a floor-bound sample is located primarily where the statistic is most heavily damped.

This also means the pooled critical value is a property of the sample as well as the items, which is worth stating wherever it is reported.

Recalibrating on each sample, which is what the function does, keeps the test close to its nominal level even when targeting is poor. **Carrying a critical value across samples does not.** Applying a value calibrated on a well-targeted sample to one sitting two logits off target gave a realised size of .028 against a nominal .05 on six items, where recalibrating on that sample held .042. Quote the value with the sample it came from, and recalibrate rather than reusing a published figure.

Can the sample dependence be removed?

Conditioning on the total score is what removed the latent distribution from the subtest problem in the companion note, and the same argument applies here. Under the null the two occasions are independent given θ and use the same items, so

$$P(r_1 = a \mid r_1 + r_2 = s) = \frac{\gamma(a) \gamma(s - a)}{\gamma_2(s)}$$

where γ_2 is the elementary symmetric function of the item set counted twice. The ratio is free of θ , verified to between 2.2e-16 and 3.9e-16 across four configurations including dichotomous and mixed-category scales. A per-respondent null is therefore available with no latent density and no plug-in person estimate.

It is not worth using. Conditioning on the total leaves few attainable pairs, so the discrete test falls short of its nominal level. Mean achievable size at $\alpha = .05$, weighted by the two-occasion score distribution of an $N(0, 1)$ sample:

Scale	strict p	mid-p	cannot reject (%)
6 items, 4 cat	0.0174	0.0493	1.4
12 items, 4 cat	0.0260	0.0524	0.2
20 items, 4 cat	0.0312	0.0526	0.0
20 items, 2 cat	0.0261	0.0341	1.2
40 items, 4 cat	0.0335	0.0482	0.0

A mid-p correction recovers the level, overshooting slightly on the middle scales. The degeneracy is minor: respondents whose total admits no rejection whatever are at most 1.4 percent of an on-target sample, because extreme totals are rare.

What settles it is that neither version improves on what the function already does. The case with no sample to pool over, and the one this construction was meant to serve, is a single respondent measured against external item parameters. Exact size given that respondent's true location, six items with four categories:

true θ	plug-in	strict	mid-p
-3	.0195	.0080	.0265

true θ	plug-in	strict	mid-p
-2	.0492	.0196	.0355
-1	.0473	.0163	.0489
0	.0428	.0184	.0540
+2	.0492	.0196	.0355
+3	.0195	.0080	.0265

The plug-in holds between .043 and .051 through the middle of the scale and turns conservative only in the tails, where the scale runs out of scores. Exact power at a true location of zero on twenty items tells the same story: at a change of one logit it is .618 for the plug-in, .626 for mid-p and .537 for strict, and mid-p's small edge tracks its slightly larger size rather than any real gain.

The negative result is worth recording because it accounts for something this note otherwise reports without explanation. The conditional null given the total does not depend on θ at all, so an error in a plug-in $\hat{\theta}$ reaches the critical value only through the weighting. That is why `conditional_crit = TRUE` performs as well as it does from a single respondent's own two estimates, and why the pooled default survives poor targeting once it is recalibrated. The default is unchanged. Worked out in `dev/conditional_total_null.R`.

A consequence for the RCI as a statistic

Because the denominator is a function of the same scores as the numerator, the RCI does not order respondents by how much they changed. A concrete pair from the six-item scale above:

	score change	change in logits	RCI
A	0 $\rightarrow$ 18	7.24	3.53
B	3 $\rightarrow$ 15	3.11	3.60

Respondent A moved the full width of the scale, 4.14 logits further than B, and scores lower on the RCI. Both are far past any critical value so nothing turns on it here, but it shows that the RCI is a standardised distance from the null rather than a measure of change, and should not be read as the latter.

The constructive version: since the critical value has to be computed from the item parameters anyway, the normalisation adds nothing that a location-specific threshold on the raw change would not. The same enumeration gives the null of $\hat{\theta}_2 - \hat{\theta}_1$ directly, so a conditional critical **change**, in logits, is available at no extra cost and is monotone in what happened. That is the same object as a conditional minimal detectable change, and connects to the score-level MDC table parked as a possible sibling function.

The no-change band in the figure

The scatter plot of occasion 1 against occasion 2 draws a band covering the changes that would not be flagged. That band was built from a smooth approximation, half-width $c \cdot \sqrt{2/I(\theta)}$, which places **both** occasions at the baseline location. The test does not do that. It uses $SE(r_1)$ and $SE(r_2)$ at two different scores.

On 300 respondents the two disagreed for 33 of them, 11 percent: 27 sat outside the drawn band without being flagged, and 6 sat inside it while flagged. The figure contradicted the table it accompanied.

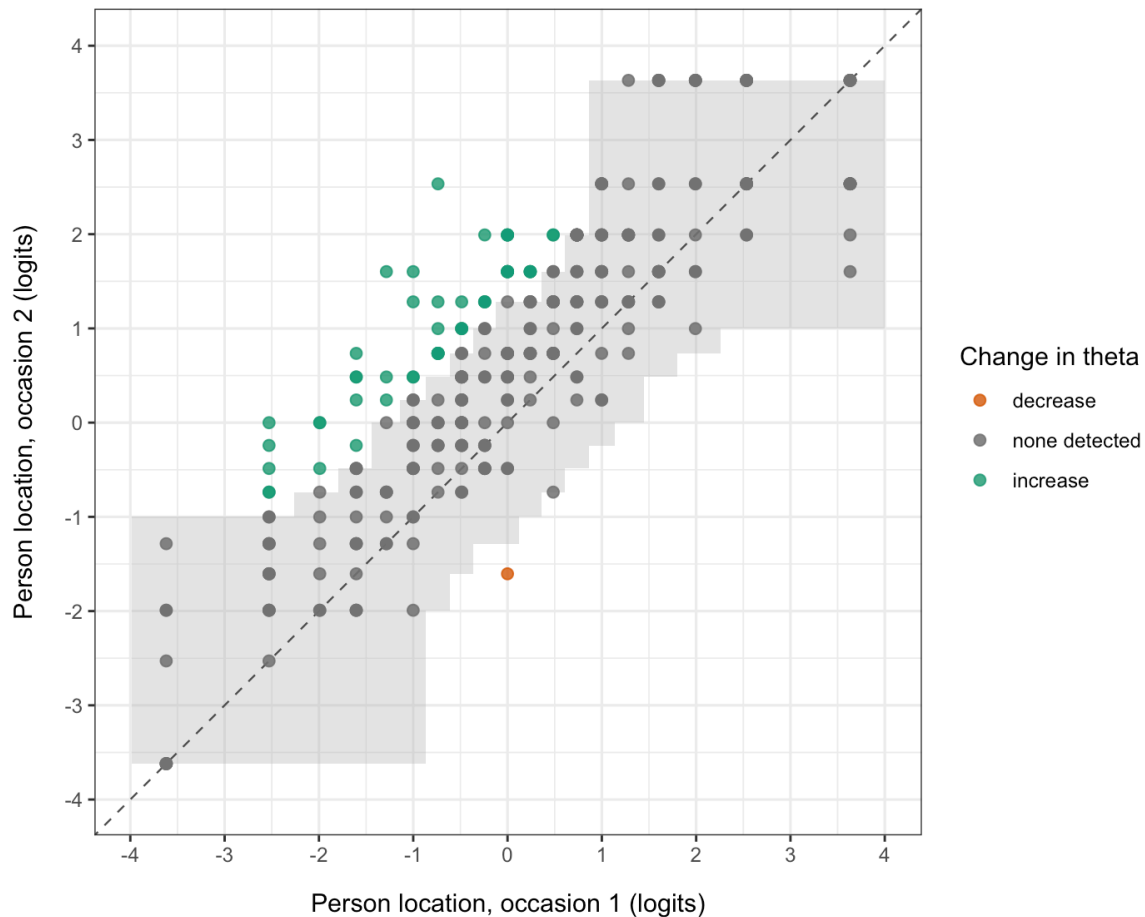
The mechanism shows in the worst case. A respondent moving from $\theta_1 = 1.23$ to $\theta_2 = 3.62$ had a band built on $SE_{diff} = 0.79$, because information is high at 1.23, while the test used 1.56, because the follow-up score is at the ceiling. The band was about half the width it should have been, in precisely the case the figure exists to display: a large change landing near a boundary.

The band is now built from the same lookup the test uses. For each attainable baseline score, the follow-up locations that would not be flagged are read off directly, so a point lies outside the band if and only if that respondent was flagged. Verified at zero disagreements out of 300 and pinned by a regression test. Because attainable scores are discrete the band is a step function, which also makes the discreteness visible rather than hiding it behind a smooth curve.

```
set.seed(1)
thr_demo <- mk(6, 4)
names(thr_demo) <- paste0("I", seq_along(thr_demo))

theta_demo <- rnorm(300, 0, 1.6)
t1 <- as.data.frame(easyRasch2::sim_partial_score(thr_demo, theta_demo))
t2 <- as.data.frame(easyRasch2::sim_partial_score(thr_demo, theta_demo + 0.7))
colnames(t1) <- colnames(t2) <- names(thr_demo)

# Item parameters are estimated from the data here, as a reader would, so that
# this figure and the reliability output below share one calibration.
RmpersonChange(t1, t2, critical = "exact", output = "ggplot")
```



Note. Null: no change beyond measurement error, $SE_diff = \sqrt{SE_1^2 + SE_2^2}$. This is the narrower of the two nulls available and does not allow for occasion-to-occasion fluctuation, so a flagged change is a necessary condition for change rather than evidence of it. Critical values computed exactly by enumerating the null, pooled across respondents: -1.72 and 1.72. Item parameters from 6 items, calibrated on both occasions stacked. Two-sided test at $\alpha = 0.05$ per respondent. 53 of 300 respondents flagged. $n = 300$ respondents.

Figure 1: Default `RMpersonChange()` figure. Three hundred respondents on a scale of six items with four response categories, generated with a true shift of 0.7 logits between occasions. The grey band covers the follow-up locations that would not be flagged from each baseline score, so a point lies outside it if and only if that respondent was flagged. Its steps are the attainable scores, and it is widest away from the centre.

The band table below gives the same object numerically.

```
thr6 <- mk(6, 4)
crit <- exact_crit(thr6, theta)

band_rows <- function(thr_list, lo, hi) {
  steps <- vapply(thr_list, length, integer(1L))
  lut <- vapply(0:sum(steps), function(r) {
    easyRasch2:::.theta_wle(
      easyRasch2:::.score_pattern(r, steps), thr_list, c(-10, 10)
    )
  }, numeric(2L))
  th <- lut[1L, ]
  se <- lut[2L, ]
  do.call(rbind, lapply(seq_along(th), function(i) {
```



```

rci <- (th - th[i]) / sqrt(se[i]^2 + se^2)
keep <- rci <= hi & rci >= lo
data.frame(baseline = th[i], lower = min(th[keep]), upper = max(th[keep]))
}))
}

b <- band_rows(thr6, crit[["lower"]], crit[["upper"]])
b$width <- b$upper - b$lower
kable(b, digits = 2, row.names = FALSE,
      col.names = c("Baseline theta", "Lower", "Upper", "Width"))

```

Baseline theta	Lower	Upper	Width
-3.73	-3.73	-1.27	2.46
-2.58	-3.73	-0.98	2.74
-2.01	-3.73	-0.72	3.01
-1.60	-3.73	-0.47	3.25
-1.27	-3.73	0.00	3.73
-0.98	-2.58	0.23	2.82
-0.72	-2.01	0.47	2.48
-0.47	-1.60	0.72	2.32
-0.23	-1.27	0.98	2.26
0.00	-1.27	1.27	2.55
0.23	-0.98	1.60	2.59
0.47	-0.72	1.60	2.32
0.72	-0.47	2.01	2.48
0.98	-0.23	2.58	2.82
1.27	0.00	3.73	3.73
1.60	0.23	3.73	3.49
2.01	0.72	3.73	3.01
2.58	0.98	3.73	2.74
3.73	1.27	3.73	2.46

The width is not monotone in distance from the centre, which is worth knowing before reading a figure. It peaks off centre, where the standard error is already large but the scale has not run out of room, and narrows again at the boundary itself because there are no further attainable scores to move to. For this scale, six items with four categories, the widest band is 3.62 logits at a baseline near ± 1.2 , against 2.47 at the centre and 2.39 at the extremes.

Two cases remain indicative rather than exact, and are documented as such: with mixed missingness the band is drawn for the most common pair of answered-item sets, and under `conditional_crit = TRUE` it uses the median critical values.

What the change figure is resting on

A change verdict is only as good as the precision behind it, and the reader of the figure above has no way to judge that from the figure alone. The same scale, six items with four response categories, summarised the two ways the package offers.

```
RMreliability(t1, draws = 200, rmu_iter = 10, seed = 1, boot = TRUE)
```

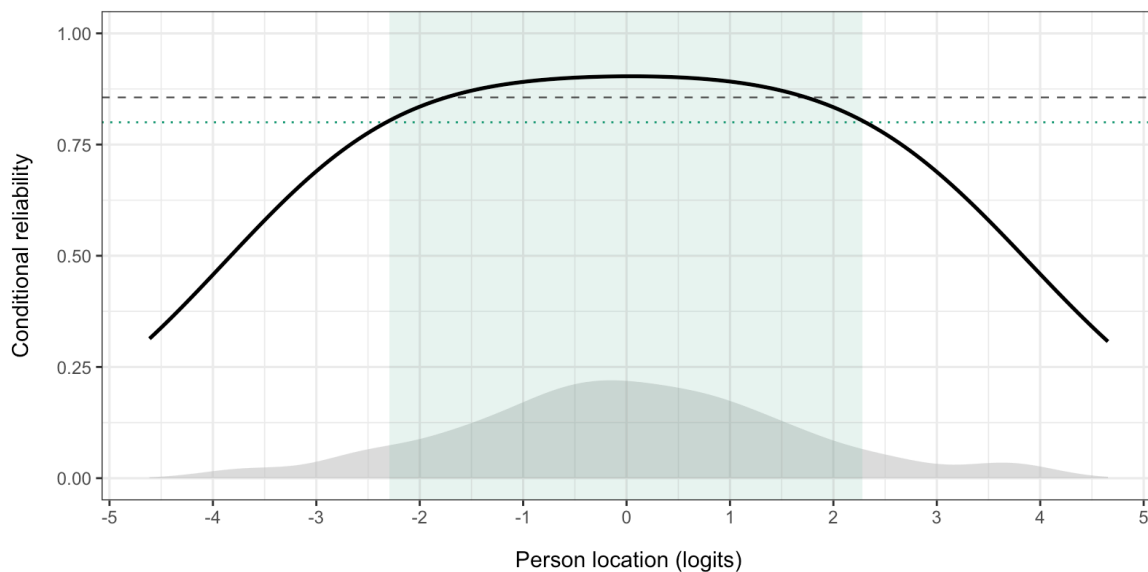
Metric	Estimate	Lower	(95% HD CI)	Upper	(95% HD CI)	Notes
Cronbach's alpha	0.868		0.850		0.889	200 bootstrap resamples
PSI	0.796		0.772		0.820	200 bootstrap resamples
Marginal (curve mean)	0.856		0.843		0.870	200 bootstrap resamples
RMU (WLE)	0.861		0.839		0.882	200 PVs, 10 RMU iterations

Marginal reliability near .85 is respectable for six items with four response categories, and the gap down to PSI is the ordinary one between a model-based coefficient integrated over the fitted latent density and a sample-based one computed from the observed spread of the estimates.

The “Marginal (curve mean)” row and the dashed line in the curve below are the same number, the latent-density-weighted mean of the conditional reliability curve. `RMreliabilityCurve()` returns it as the `marginal_ratio` attribute, and the two agree to machine precision whenever both are given the same data and the same calibration.

A single coefficient hides where that precision lies, which is the same point the stepped band makes in the change figure.

```
RMreliabilityCurve(t1, statistic = "reliability", benchmark = 0.8)
```



Note. Conditional reliability, $\sigma^2 / (\sigma^2 + \text{SEM}(\theta)^2)$ with $\sigma = 1.54$, from test information summed over 6 items using CML thresholds. The curve is a property of the items, not of the sample. Respondent locations by WLE. Shaded distribution shows where respondents fall. Dashed line is the marginal reliability (0.856), the flat summary a single coefficient reports. Dotted line and shaded band mark reliability ≥ 0.80 , reached by 83% of respondents. Information-based standard errors are asymptotic and only approximate for a scale this short. $n = 300$ respondents.

Figure 2: Conditional reliability across the scale for the same six items with four response categories, with the region reaching .8 shaded and the respondent distribution behind it.

Read Figure 1 and 2 together. Where this curve is high the band in the change figure is narrow and a modest change is detectable. Where it falls away the band widens and only a large change clears it. The band narrowing again at the very extremes is not precision returning, it is the scale running out of scores.

Response dependence between occasions

Everything above assumes the two occasions are independent given θ . That is a conditional statement, and it is compatible with the two estimates being strongly correlated marginally, which they are. Both measure the same person, and that shared θ is what makes their difference interpretable. The assumption concerns what is left once θ is accounted for.

What it rules out is carry-over: a time-2 answer depending on the time-1 answer to the same item beyond what θ explains. Administering the occasions separately does not deliver conditional independence on its own. Marais [1] models the failure by shifting the time-2 difficulty of item j by $(1 - 2x_{j1})d$, so a positive response at time 1 makes that item easier at time 2 and a negative one makes it harder. Marais and Andrich [2] separate this **response dependence** from **trait dependence**, which is multidimensionality. Only the second is a failure of unidimensionality, and the two call for different remedies.

This matters here in a way it does not for the companion note on the PCA/t-test protocol. There, conditional independence between the two item subsets is the null hypothesis, so a failure of it is what the test exists to detect. Here it is an auxiliary assumption: the null is that θ did not change, and carry-over can be present while θ is fixed.

```
# Marais's model, generalised to polytomous items by shifting every threshold
# of item j by (1 - 2 * x_j1 / m_j) * d, which reduces to her dichotomous form
# when m_j = 1. Shifting all thresholds by s is the same as shifting theta by
# -s, so the shift is applied as an effective theta.
rpcm <- function(theta_eff, thr) {
  m <- length(thr)
  num <- matrix(0, length(theta_eff), m + 1L); cs <- cumsum(thr)
  for (k in seq_len(m)) num[, k + 1L] <- k * theta_eff - cs[k]
  num <- exp(num - apply(num, 1L, max))
  cp <- t(apply(num / rowSums(num), 1L, cumsum))
  max.col(cp >= runif(length(theta_eff)), ties.method = "first") - 1L
}

sim_pair <- function(thr_list, theta1, theta2, d = 0) {
  k <- length(thr_list); n <- length(theta1)
  mj <- vapply(thr_list, length, integer(1L))
  x1 <- vapply(seq_len(k), function(j) rpcm(theta1, thr_list[[j]]), integer(n))
  sh <- if (d == 0) matrix(0, n, k) else (1 - 2 * sweep(x1, 2L, mj, "/")) * d
  x2 <- vapply(seq_len(k), function(j)
    rpcm(theta2 - sh[, j], thr_list[[j]]), integer(n))
  list(t1 = x1, t2 = x2)
}

# Wider item spread than mk() above, following Marais's [-2.5, 2.5]
mk_wide <- function(k, ncat) {
  lapply(seq(-2.5, 2.5, length.out = k),
    function(b) b + seq(-0.9, 0.9, length.out = ncat - 1L))
}

score_lut <- function(thr_list) {
  steps <- vapply(thr_list, length, integer(1L))
  vapply(0:sum(steps), function(r)
    easyRasch2:::theta_wle(easyRasch2:::score_pattern(r, steps),
      thr_list, c(-10, 10)), numeric(2L))[1L, ]
}
```

The mechanism is easiest to see with no true change at all. Holding every respondent at one location and generating both occasions from it, the mean observed change should be zero at every location, and is when $d = 0$. Twenty items, four categories, item locations over $[-2.5, 2.5]$, 4000 respondents per cell, change in logits.

```
set.seed(808)
thr_d <- mk_wide(20, 4); th_lut <- score_lut(thr_d)
grid_t <- seq(-2.5, 2.5, by = 0.5)
drift <- sapply(c(0, 0.5, 1, 1.5), function(d)
  sapply(grid_t, function(t0) {
    p <- sim_pair(thr_d, rep(t0, 4000), rep(t0, 4000), d = d)
    mean(th_lut[rowSums(p$t2) + 1L] - th_lut[rowSums(p$t1) + 1L])
  }))
dimnames(drift) <- list(sprintf("%.1f", grid_t),
  paste0("d = ", c(0, 0.5, 1, 1.5)))
kable(drift, digits = 3)
```

	d = 0	d = 0.5	d = 1	d = 1.5
-2.5	-0.007	-0.210	-0.359	-0.465
-2.0	-0.007	-0.136	-0.227	-0.294
-1.5	0.000	-0.080	-0.136	-0.179
-1.0	0.002	-0.048	-0.079	-0.100
-0.5	0.001	-0.020	-0.037	-0.045
0.0	0.005	-0.002	-0.017	0.003
0.5	0.003	0.026	0.043	0.047
1.0	0.004	0.047	0.080	0.106
1.5	-0.004	0.097	0.148	0.185
2.0	0.001	0.137	0.231	0.293
2.5	0.000	0.200	0.361	0.459

Carry-over produces an antisymmetric **outward drift**: zero at the item mean, pushing respondents further out the further out they already are, and reaching almost half a logit at the ends by $d = 1.5$. This is the individual-level mechanism behind the group-level result Marais reports [1], where easy items put the sample above the item mean, everyone drifts up, and a group mean shows change that did not happen.

What it does to the test

The remaining results come from `dev/carryover_dependence.R`, which is not run at render time. That script also validates the generator against Marais's published Table 4, reproducing her mean same-item residual correlations of $-0.01, 0.14, 0.31$ and 0.47 as $-0.02, 0.13, 0.28$ and 0.44 .

With no true change, and critical values calibrated under independence as this note recommends, the test becomes conservative rather than liberal. The standard deviation of the RCI falls from about 0.97 to between 0.63 and 0.73, so the exact critical values are too wide for the null actually in force. Type I error against a nominal .05, on-target sample:

Scale	d = 0	d = 0.5	d = 1	d = 1.5
20 items, 4 cat	.0487	.0267	.0133	.0074
6 items, 4 cat	.0509	.0261	.0121	.0056

Scale	$d = 0$	$d = 0.5$	$d = 1$	$d = 1.5$
25 items, 2 cat	.0488	.0262	.0103	.0032

The direction of the false positives is the part that does damage. At $d = 0$ they split evenly, as they must. As dependence grows they polarise, because the drift carries each respondent away from the item mean and the sign of that movement is fixed by where the respondent already was. Twenty items with four categories, on target:

d	share “increase” among those above the items	share “decrease” among those below
0	0.51	0.49
0.5	0.75	0.75
1.0	0.88	0.89
1.5	0.97	1.00

With a real change the same drift manufactures a severity gradient. Simulating a uniform improvement of 0.5 logits for every respondent on twenty items with four categories, the estimated change in logits comes out as:

d	overall	below items	near items	above items	above - below
0	0.498	0.497	0.498	0.496	-0.001
0.5	0.436	0.349	0.431	0.544	0.195
1.0	0.374	0.225	0.368	0.551	0.326
1.5	0.324	0.138	0.316	0.547	0.409

A flat treatment effect reads as a 0.41 logit gradient in baseline severity, and a reader with no access to d has no way to tell that apart from a real interaction between treatment and baseline. Power falls at the same time, from .189 to .043 on twenty items and from .083 to .013 on six.

Why `null = "retest"` does not help

`retest_sd` adds an independent occasion deviation at each administration. That is symmetric extra variance, and it widens an already conservative null. The drift above is an antisymmetric shift in the conditional mean, which extra variance does not touch. The two estimands in `RMpersonChange()` are both defined under conditional independence, so neither covers this.

Detecting it

Racking the two occasions into one analysis of $2k$ items and screening the same-item pairs against `RMlocdepQ3Cutoff()` finds it well below the level that does damage. Twenty items, four categories, $n = 500$:

d	pairs flagged of 20	runs flagging any
0	0.0	0%
0.5	3.3	100%
1.0	14.2	100%

Marais [1] judged $d = 0.5$ likely to pass unnoticed and $d = 1$ possibly so, working from raw residual correlations and from differential item functioning by occasion, neither of which carried a calibrated reference. Against a simulation-based cutoff the lowest level she studied is caught in all 25 runs, with

a clean zero under the null. Her conclusion about detectability reflects the diagnostics available at the time rather than a limit of the data.

Two details decide whether this works in practice. **The statistic has to be centred on the cross-occasion pairs**, not Yen-corrected against every pair. Subtracting the mean of all off-diagonal $Q3$ mixes within-occasion pairs into the reference and costs power: at $d = 0.5$ the two centrings detect in 96 and 100 percent of runs respectively, and the gap widens when real change is also present.

Heterogeneous true change does not masquerade as carry-over, which was the objection worth settling before recommending the screen at all. Both act on the cross-occasion block, so it is not obvious on paper that they separate. They do, because carry-over is item-specific and change is not:

condition	same-item	other cross-occasion	contrast	within-occasion
no change, no carry-over	-0.004	0.000	-0.004	0.000
uniform change +0.5	-0.004	0.000	-0.004	0.000
change SD 0.6	-0.039	-0.034	-0.005	0.036
carry-over $d = 0.5$	0.147	-0.005	0.152	-0.004
carry-over $d = 1.0$	0.274	-0.008	0.281	-0.007
both, $d = 1$ and SD 0.6	0.224	-0.032	0.255	0.020

Uniform change is absorbed into the time-2 item locations and leaves no trace, which is the property Marais relies on for the rack analysis. Change that varies between respondents is a second person dimension: it lifts the within-occasion pairs, depresses the cross-occasion block as a whole, and leaves the contrast flat. Carry-over lifts the same-item pairs and nothing else. When both are present the two signatures add without interfering. Screened against the simulated cutoff, no run out of five flagged a single pair under heterogeneous change at SD 0.6 or SD 1.0, so the screen does not confuse the two.

Scope and limits

- **Complete data only.** Missing data breaks sufficiency of a single sum score. Each distinct missingness pattern has its own score distribution and its own lookup, so the enumeration still works but per pattern.
- **null = "retest" needs quadrature.** The occasion deviations are continuous, so the joint becomes a double integral over u_1, u_2 rather than a pure sum. Still deterministic and cheap at, say, 21 nodes a side.
- **EAP is fine.** On complete data an EAP estimate is also a function of the sum score.
- Validated here for WLE with CML thresholds. Not yet checked against MML item parameters.
- **Conditional independence between occasions is assumed, not tested.** The enumeration is correct under it and says nothing about carry-over. See the section above, and screen for it before relying on a change verdict.

Implemented

Shipped as `critical = "exact"`, now the default in `RmpersonChange()`, in `R/person_change_exact.R`. Measured against a 4000-iteration simulation on the same data: critical values agree to within Monte Carlo error, p-values correlate at 0.9998, classifications are identical, and the exact path takes 0.08 s against 13.7 s. Incomplete data is enumerated within each missingness pattern, `null = "retest"` integrates the occasion deviations out by quadrature (they are independent, so the joint factorises and the retest null costs no more than the measurement null), and EAP is supported by holding the prior at the one the observed respondents were scored under.

Having an exact reference immediately paid for itself twice. It exposed a bug in the simulation path, which had been re-estimating the EAP prior from each null dataset rather than holding it at the observed one. That is now fixed, and the two paths agree to four decimals under EAP. It also supplied the lookup needed to fix the figure, described in the next section.

It also removes the one real objection to `conditional_crit = TRUE`: a per-respondent critical value from simulation rests on `sim_iter` draws, roughly 25 of which land in each tail at $\alpha = .05$, whereas $P(r \mid \theta_i)$ is available per respondent for free.

It would also improve the planned simulation study, whose first aim is the realised Type I error of the simulated critical value. Much of the noise in that comparison disappears if the reference is exact.

Reproduction

Scratch scripts used for the simulated column are not part of the package. The exact computation is self-contained in the `exact-fns` chunk above and needs only `easyRasch2:::pcm_cat_probs()`, `:::theta_wle()` and `:::score_pattern()`.

Bibliography

- [1] I. Marais, “Response Dependence and the Measurement of Change,” *Journal of Applied Measurement*, vol. 10, no. 1, pp. 17–29, 2009.
- [2] I. Marais and D. Andrich, “Formalizing Dimension and Response Violations of Local Independence in the Unidimensional Rasch Model,” *Journal of Applied Measurement*, vol. 9, pp. 200–215, 2008.